Fifth Semester B. Tech. (Electronics and Communication Engineering) Examination

WIRELESS COMMUNICATION

Time : 3 Hours] [Max. Marks : 60

Instructions to Candidates :—

(1) Assume suitable data wherever necessary.

(2) Illustrate your answers with neat sketches, wherever necessary.

(3) Choice is available in Question No. **1** [Solve **1. (a)** and **1. (b)** **OR** **1. (c)** and **1. (d)**].

(4) All other questions are compulsory.

1. (a) Explain Hand off strategies in Wireless Communication System. 5(CO1,2)

(b) Consider a single high Power transmitter in Airtel Network that can support **40** voice channels over an area of **140 (Km)^2**. With the available spectrum. If this area is equally divided into seven smaller areas (cells), each supported by lower power transmitter so that each cell supports **30%** of the channels, then determine :

(i) Coverage area of each cell.

(ii) Total no. of voice channels available in cellular system.

Comment on the result obtained. 5(CO1,2)

OR

(c) Explain Frequency Reuse concept in Wireless Communication System. Illustrate relationship between cluster size **M** and system capacity **C** with the help of Frequency Reuse concept. 5(CO1,2)

(d) Elaborate Co channel Interference with its reduction methods. 5(CO1,2)

2. (a) Elaborate the concept of multipath Fading with the help of diagram. 5(CO2,3)

(b) In a cell of Reliance Jio network with **100** mobile stations, on an average **30** requests are generated during an hour. With average holding time **T = 360** seconds. Calculate arrival rate $\boldsymbol{\lambda}$ and offered load **a**. 5(CO2,3)

3. (a) Define the term Diversity. Explain its significance in mobile radio environment. 5(CO2,3)

(b) Emphasize the importance of Quadrature Amplitude Modulation (QAM) in Wireless Communication. 5(CO3)

4. (a) Spread Spectrum is recommended for multipath propagation, Justify. 5(CO3)

(b) In TDMA frame structure, each frame consists of eight time slots and each time slot contains **156·25 bits** and data is transmitted at **270·833 kbps** in the channel, find :

(a) The time duration of a bit.

(b) The time duration of a slot.

(c) The time duration of a frame.

(d) How long must a user occupying a single time slot wait between two successive transmissions ? 5(CO3)

5. (a) Explain the sequence of steps performed in GSM architecture to establish a call from one mobile user to another mobile user that belongs to different MSC. Use the block schematic of GSM architecture to explain above steps. 10(CO4)

6. (a) Consider a Airtel Network cell with **S = 2** channels with 100 Mobile stations generating call requests on an average **30 requests / hr**. Average Holding time **360 seconds**. Calculate load a, no. of re-routed calls and efficiency of the system. (**Blocking Probability B = 0·53**). 5(CO3,4)

(b) What is the principle of OFDM systems ? Explain its operation with neat block diagram. 5(CO3)